

Research Assistant — Final Project Report

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Overview

The goal of this project was to develop an AI-based Research Assistant that helps users find relevant scholarly articles. The assistant is built using a combination of data science, network analysis, clustering, regression, and modern NLP techniques. Our system leverages both an original dataset (articles published up to 2017) and additional data scraped from online sources (crawled data) that are used for testing. All source code has been documented and uploaded to GitHub, with collaborators added as required.

The project is divided into several key parts:

Data Crawling: In previous phases, we scraped additional scholarly data. This crawled data has been preserved and later used as testing data.

Exploratory Data Analysis (EDA): We perform both basic EDA (bar charts, correlation analysis, wordclouds) and network analysis (citation network, co-authorship network, venue network, and their temporal evolution) to extract useful insights from the data.

Data Extrapolation via Clustering: We detect communities in the author–author network using multiple clustering algorithms, evaluate them, and then aggregate and name these communities by extracting keywords from paper texts. We also perform paper–paper clustering based on embeddings that incorporate community keywords.

Citation Regressor: We build regression models to predict the number of citations a paper will receive using textual features (title and abstract), publication year, and (as a bonus) features derived from the author–author network.

Product – Research Assistant: Finally, we integrate the above components into a product. Our Research Assistant uses a retrieval-augmented generation (RAG) approach to search a vector database of paper embeddings. The assistant can retrieve relevant papers based on topic queries (enhanced by KeyBERT for keyword extraction) or author queries (using spaCy NER). In addition, a bonus module uses an LLM (via the OpenAI API) to generate summary reports of the retrieved papers.

Below, we describe each part in detail and provide representative code blocks.

0. Crawling

Note: This phase was completed in a previous project phase. The crawled data have been preserved and are later combined with our original dataset for testing.

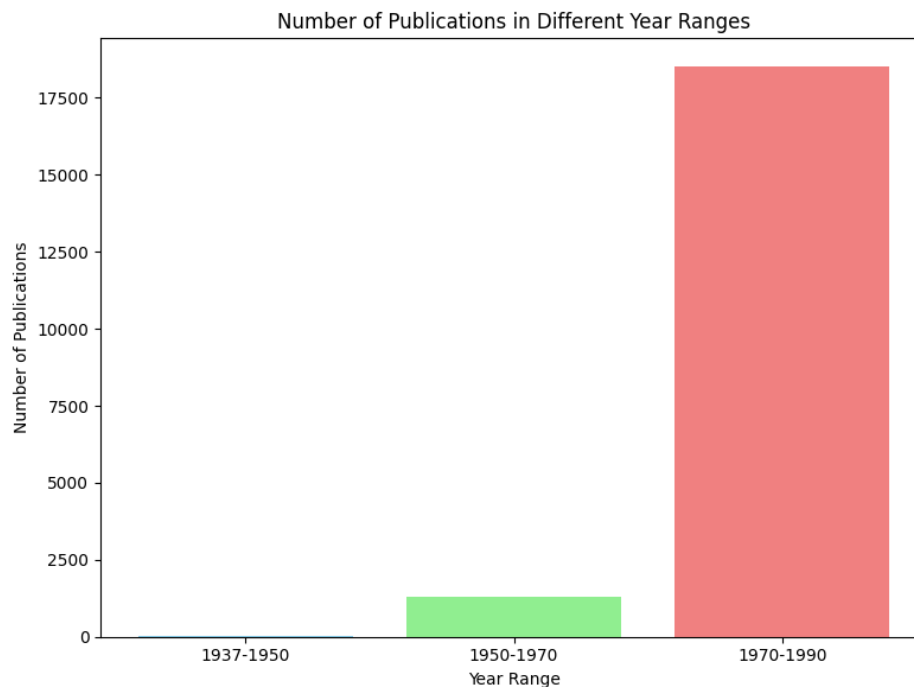
1. Exploratory Data Analysis (EDA)

1.1 Basic EDA

Publication Trends:

We created bar charts showing the number of publications over three year ranges (1937–1950, 1950–1970, 1970–1990) to compare research activity over time.

Fig. 1.1: Bar chart of number of publications in a given year



References and Authors:

Bar charts were produced to visualize the number of references and the number of authors per year. We computed Pearson and Spearman correlation coefficients to examine the relationships between the number of authors and both references and citations.

Fig 1.2: bar chart of the number of references over the years

Fig. 1.3: bar chart of the number of authors over the years

Title Length and Abstract Analysis:

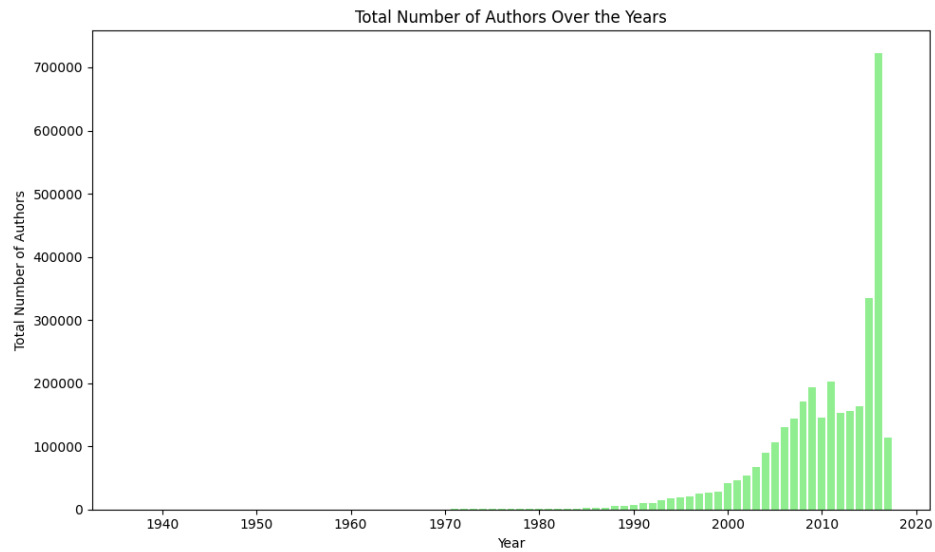
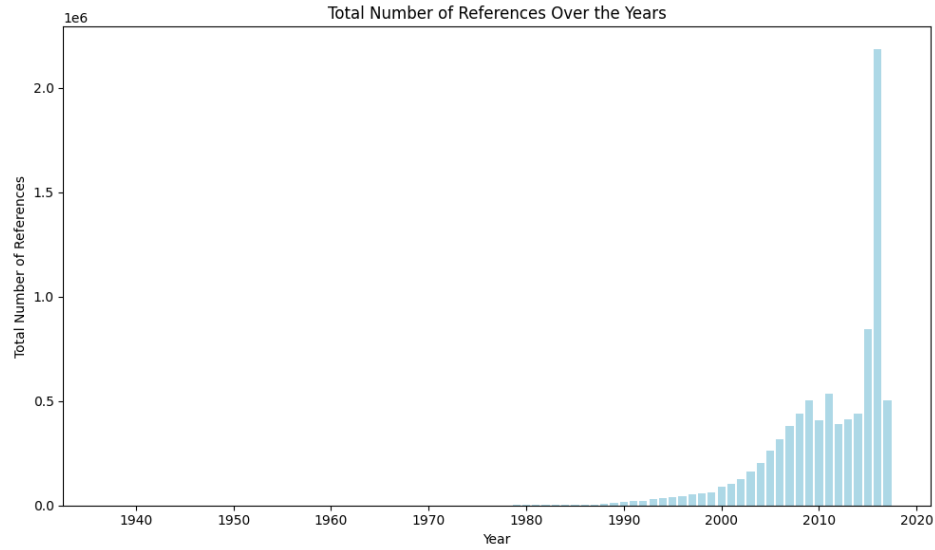
We drew a bar chart of title lengths over the years, generated a wordcloud from abstracts, and computed correlations between the title lengths of papers and those of the papers they reference.

Fig. 1.4: bar chart of the title length over the years

Fig. 1.5: wordcloud of the abstracts.

Top Entities:

We identified the top 10 authors by publication count and citations, as well as the top 10 papers by the number



of references and citations.

Fig. 1.6: top 10 authors with the most publications.

Fig. 1.7: top 10 authors with the most citations

Fig. 1.8: top 10 papers with the most citations within the dataset.

Fig 1.9: top 10 papers with the most references.

Predictive Analysis:

We explored how well the number of publications can predict the number of citations for a given author using regression analysis.

Fig. 1.10: predict the number of citations for a given author.

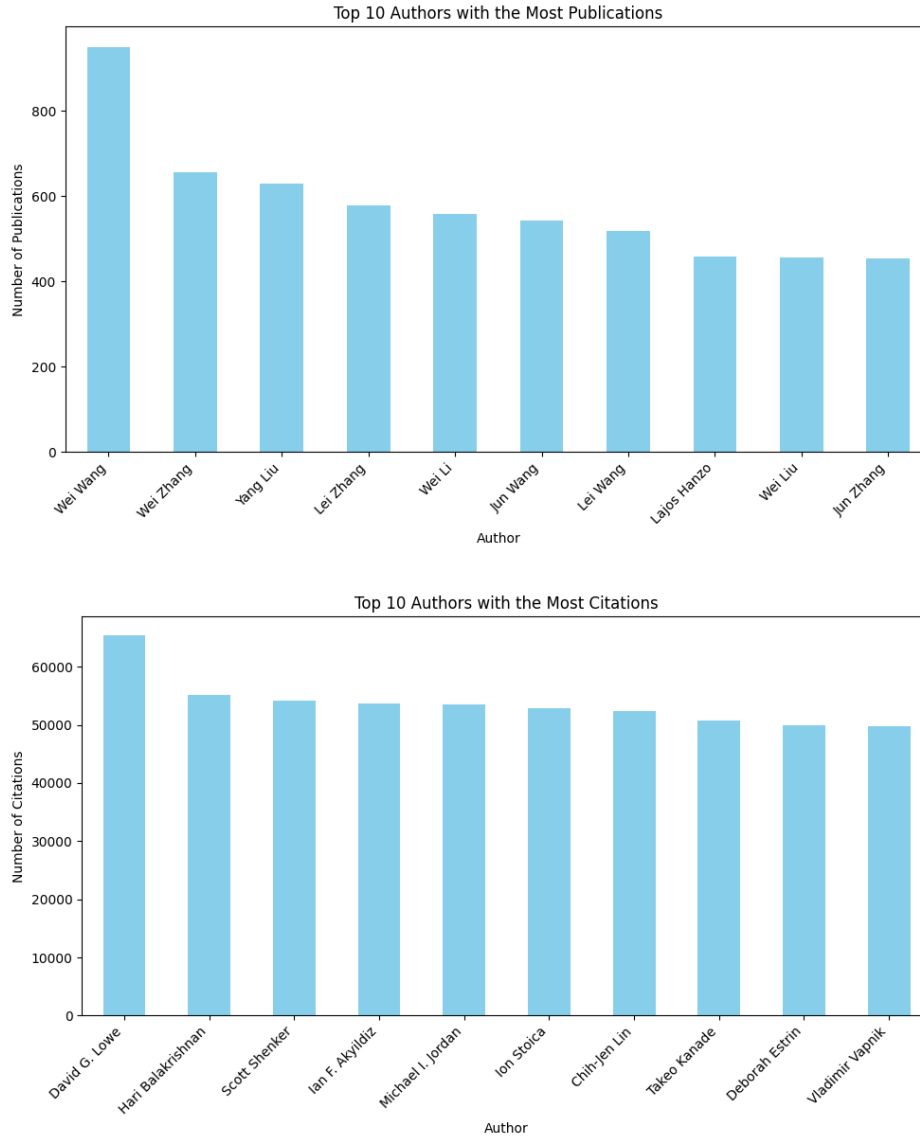


Fig. 1.14: Top 10 Largest Author Communities

Venue Network:

A network of venues (conferences/journals) was created by linking venues that have inter-citation relationships, enabling analysis of interdisciplinary collaborations and emerging fields.

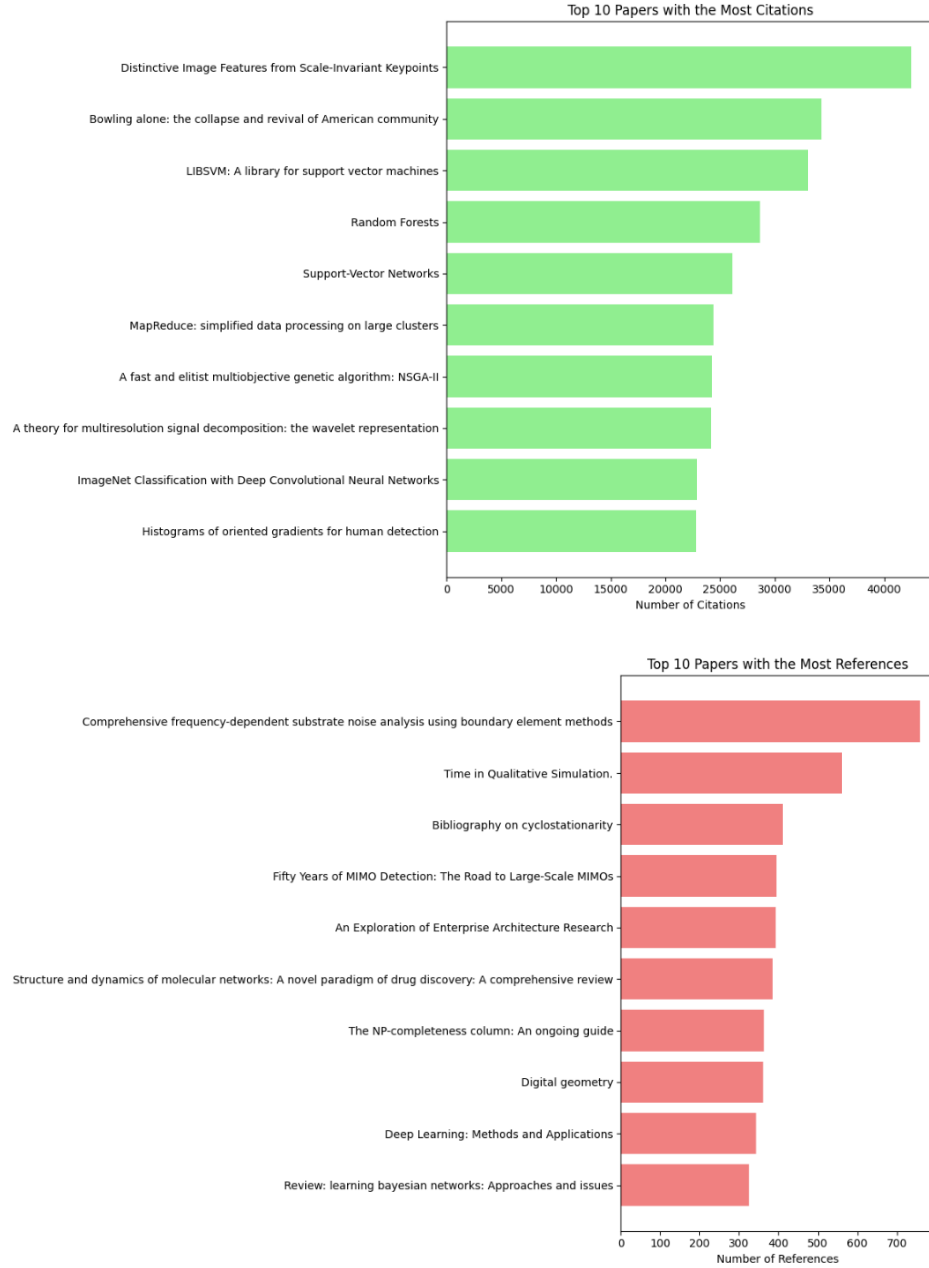
Temporal Evolution:

We constructed citation networks for different years to observe the network's evolution, identify bursts of influential papers, and analyze how new papers integrate into the existing network.

Figure 1 Fig. 1.15: Temporal Evolution of the Citation Network

Fig. 1.16: Citation Count vs. Publication Year

Fig. 1.17: Temporal Evolution of New Paper Integration into the Existing Network



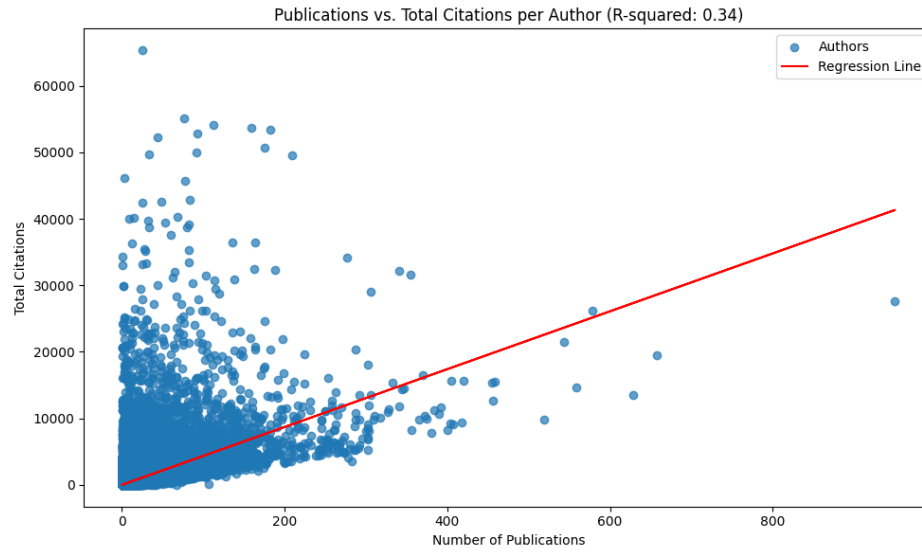
2. Data Extrapolation via Clustering

2.1 Community Detection

Author–Author Network:

We built an author–author network where each author node retained data about their published papers. Three clustering algorithms (e.g., Louvain, Label Propagation, and Girvan–Newman) were applied to detect communities. We then evaluated these clusters using the clustering coefficient and selected the best-performing algorithm.

Fig. 2.1: Author–Author Network Summary:



Top 10 influential papers using PageRank:

Paper ID	PageRank Score
c1b6b493-01ef-420f-be44-7bacfe34e846	0.0009
b944f77f-113b-4a02-ae5e-d4a124b8fd5b	0.0008
f6bd8b64-684d-429a-aab5-8ff3a2c23cd6	0.0007
65d5ccdc-7022-45b0-adf9-0385273b1283	0.0005
d3e00e7e-1c64-4d7a-b2b2-1ad98ba4c706	0.0005
8026f56a-a93e-4933-8ead-c9aa9e3f0498	0.0004
50dd56db-151d-4d62-8576-65f0ef6f381b	0.0004
748a2ab3-8b5f-4d0a-9e2d-af685089843a	0.0004
e2f7a74a-8430-4463-94ce-fe85dfd309f9	0.0004
62549bc2-e0b3-46e8-8d32-390dded105d5	0.0004

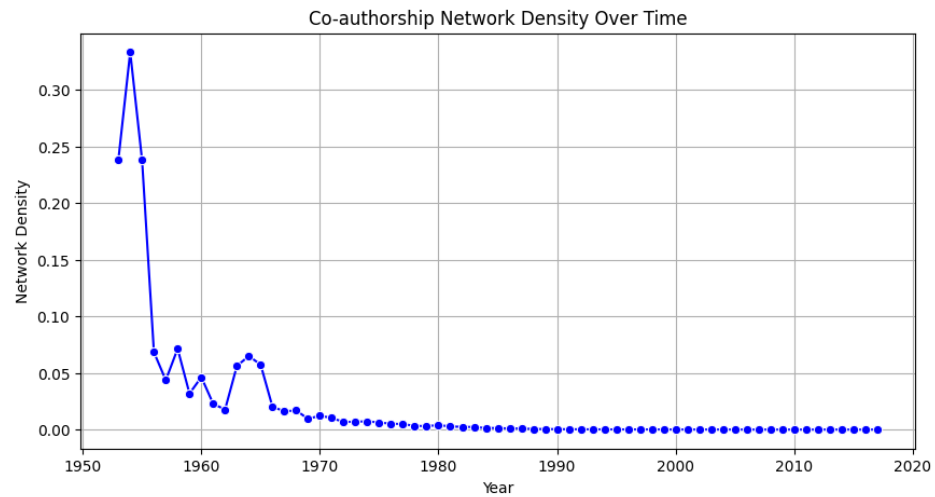
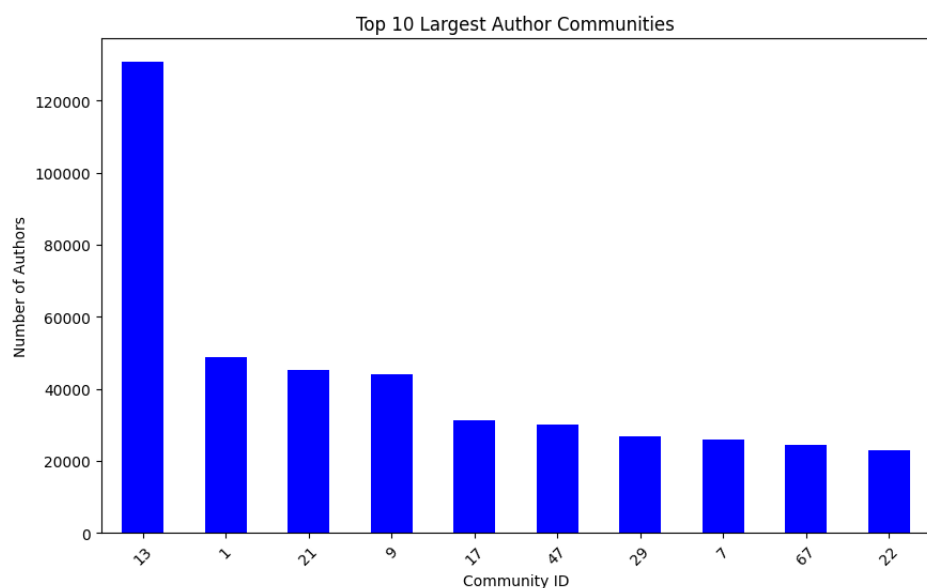
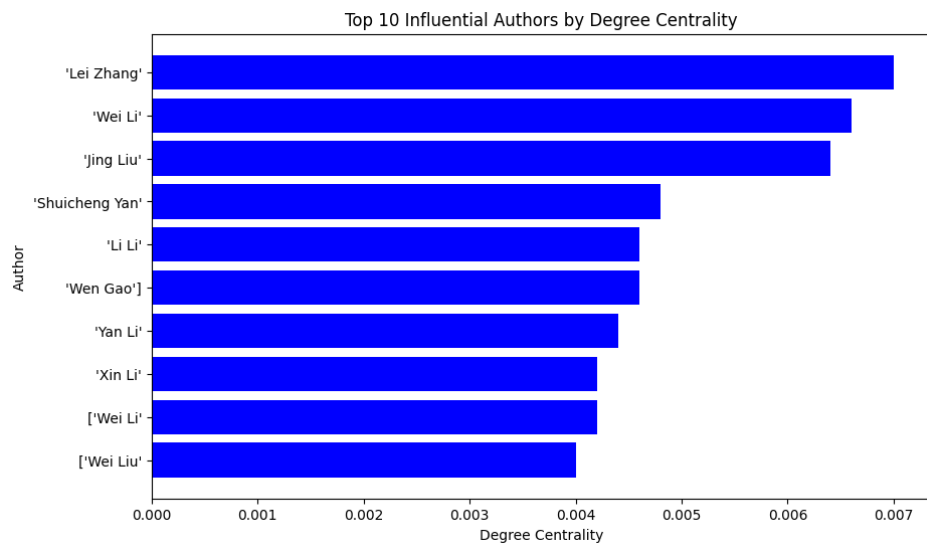


Fig 2.2: author communities

Fig. 2.3: clustering algorithm

Naming Communities:



For each paper, keywords were extracted from its title and abstract using KeyBERT. Each paper was associated with the communities of its authors, and the keywords from all papers within a community were aggregated using frequency counts. Communities were then named by selecting the most frequent keyword, which was assumed to best represent the dominant research area of that community.

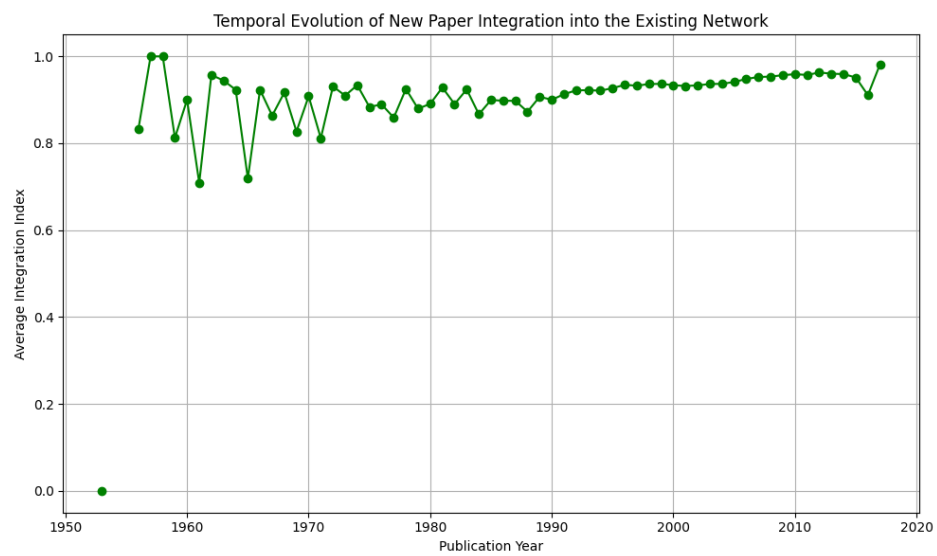
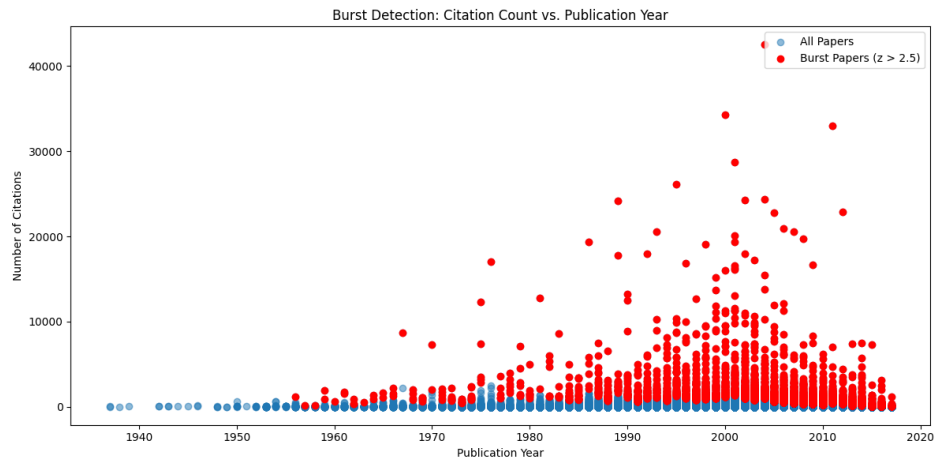
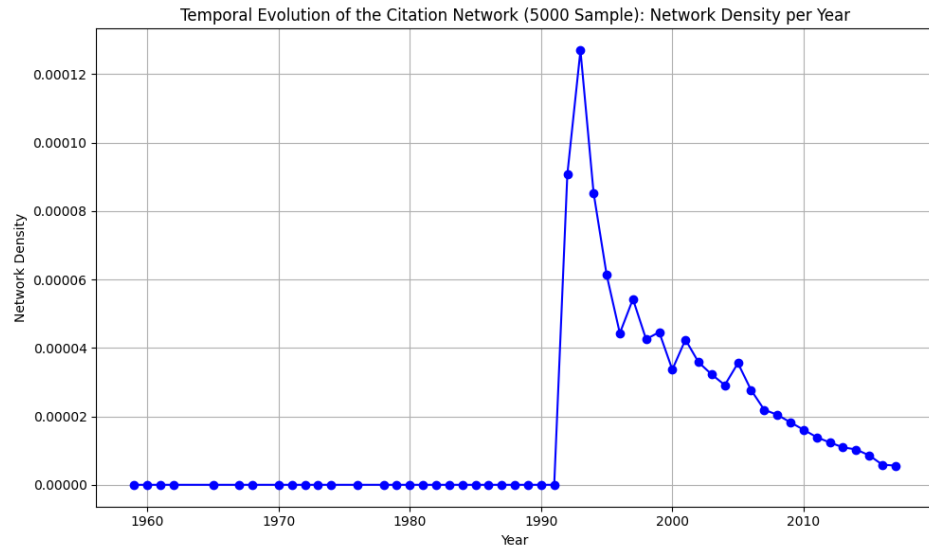
Fig. 2.5: Sample Keywords

Fig. 2.6: Author–Author Network

2.2 Paper-Paper Clustering via Embedding

Embedding and Clustering:

Each paper’s abstract and title were embedded using SentenceTransformer. The paper embedding was augmented by appending the community keywords as plaintext. A clustering algorithm (e.g., KMeans) was then



```
Author–Author Network Summary:
Number of nodes (authors): 957242
Number of edges (co-authorship relationships): 3814744
```

```
Community 770 (size: 2): ['Cláudia Antunes', 'Pedro Terras Crespo']
Community 771 (size: 5): ['Dale Schuurmans', 'Martha White', 'Ruitong Huang', 'Xinhua Zhang', 'Yaoliang']
Community 772 (size: 3): ['Atalay Özgövde', 'Giuseppe De Giacomo', 'Yves Lespérance']
Community 773 (size: 2): ['Insung Ihm', 'Jung-Heum Park']
Community 774 (size: 2): ['Divesh Srivastava', 'Peter G. Selfridge']
Community 775 (size: 2): ['Matteo Ciman', 'Ombretta Gaggi']
Community 776 (size: 8): ['David Shun-Hill Wong', 'Jizheng Chu', 'Pin-Ho Lin', 'Po-Feng Tsai', 'Shi-Jer']
Community 777 (size: 2): ['José Hernández-Orallo', 'M. José Ramírez-Quintana']
Community 778 (size: 3): ['Andreas Ulrich', 'Andrej Pietschker', 'Peter Zimmerer']
Community 779 (size: 4): ['Bayan S. Sharif', 'Charalampos C. Tsimenidis', 'O.R. Hinton', 'S.J. Yi']
Community 780 (size: 3): ['Lei Jin', 'Xiaojun Ye', 'Yawei Zhang']
Community 781 (size: 2): ['Jiaguo Qi', 'Yasuyo Makido']
Community 782 (size: 3): ['Carolina Bernardos', 'Miguel Mediavilla', 'Sandra Martínez']
Community 783 (size: 4): ['A. S. Al-Fhaid', 'Eman Alaidarous', 'Fayyaz Ahmad', 'Malik Zaka Ullah']
Community 784 (size: 2): ['Giuliano Antoniol', 'Martin Pinzger']
Community 785 (size: 1): ['David H. Ackley']
Community 786 (size: 1): ['Sabrina Ahmad']
Community 787 (size: 2): ['Arshad Hussain', 'Muhammad Ahmad Raza']
Community 788 (size: 5): ['Nordin Saad', 'Rosdiazli Ibrahim', 'Sabo Miya Hassan', 'Tran Duc Chung', 'Vij']
Community 789 (size: 2): ['Daniel T. Fitzpatrick', 'Martin E. Newell']
Community 790 (size: 2): ['Alexander Maedche', 'Peter Weiß']
Community 791 (size: 4): ['Madanlal Musuvathi', 'Todd Mytkowicz', 'Xipeng Shen', 'Yufei Ding']
Community 792 (size: 4): ['Andreas Rausch', 'Christian Bartelt', 'Martin Vogel', 'Tim Warnecke']
Community 793 (size: 3): ['Cesare Rossi', 'F. Penta', 'Sergio Savino']
Community 794 (size: 1): ['Andy Way']
Community 795 (size: 3): ['Hideo Masuda', 'Takumi Kouga', 'Yoshiko Ishibashi']
Community 796 (size: 3): ['Amirhossein Amiri', 'Armaghan Azhdari', 'Mehdi Koosha']
Community 797 (size: 2): ['Makoto Suzuki', 'Shigeichi Hirasawa']
Community 798 (size: 3): ['Kurt Rothermel', 'Sven Krause', 'Thomas Magedanz']
Community 799 (size: 2): ['Muhammad Sheraz Arshad Malik', 'Suziah Sulaiman']
Community 800 (size: 1): ['Markus Fockel']
Community 801 (size: 2): ['Collin G. DeYoung', 'Tanja Sauer']
```

```
Greedy Modularity found 976 communities.
Label Propagation found 984 communities.
Girvan-Newman (first level) found 977 communities.

Average Clustering Coefficient for each clustering algorithm:
Greedy Modularity: 0.7770
Label Propagation: 0.7757
Girvan-Newman: 0.7771

Best clustering algorithm based on average clustering coefficient:
Girvan-Newman with an average clustering coefficient of 0.7771
```

applied to group papers based on thematic similarity.

Fig. 2.7: Embed the abstract

Evaluation:

The clustering performance was evaluated using the Davies–Bouldin Index (DBI) and Silhouette scores. Additionally, the unique venues of the dataset were reported, and the Jaccard Similarity Index between our clustering and the clustering based on venue labels was computed.

```

id \
987231 3e5ef05b-bb89-4717-bfc3-74a467529ded
79954 653894e3-b581-412c-ae3f-71d267b0ea9d
567130 0ba8b19f-9659-453f-8deb-50b87e26e41f
500891 f12c84ba-82a9-4835-b5fa-5e6715b1b6b3
55399 5d197014-1fbb-40d8-8223-378dea0d4a30

title \
987231 Slice-based parallelization in HEVC encoding: ...
79954 MIDAS: A Middleware for Information Systems wi...
567130 Automatically controlled pan-tilt smart camera...
500891 Particle Filter for Visual Tracking Using Mult...
55399 A reference model for crowdsourcing as a service

keywords
987231 [efficiency video, hdtv uhdtv, video coding, h...
79954 [database increases, locks required, difficult...
567130 [smart camera, algorithm camera, innovative re...
500891 [cameras optimally, data fusion, results algor...
55399 [online crowdsourcing, crowdsourcing applicati...

```

```

Built Author-Author Network from 100-sample.
Number of nodes: 318 Number of edges: 450
Detected 100 communities.
=== Sample Paper Communities ===

```

```

id \
987231 3e5ef05b-bb89-4717-bfc3-74a467529ded
79954 653894e3-b581-412c-ae3f-71d267b0ea9d
567130 0ba8b19f-9659-453f-8deb-50b87e26e41f
500891 f12c84ba-82a9-4835-b5fa-5e6715b1b6b3
55399 5d197014-1fbb-40d8-8223-378dea0d4a30

title communities
987231 Slice-based parallelization in HEVC encoding: ... [3]
79954 MIDAS: A Middleware for Information Systems wi... [60]
567130 Automatically controlled pan-tilt smart camera... [61]
500891 Particle Filter for Visual Tracking Using Mult... [33]
55399 A reference model for crowdsourcing as a service [34]

```

```

modules.json: 100% ██████████ 349/349 [00:00<00:00, 15.7kB/s]
config_sentence_transformers.json: 100% ██████████ 116/116 [00:00<00:00, 7.61kB/s]
README.md: 100% ██████████ 10.7k/10.7k [00:00<00:00, 736kB/s]
sentence_bert_config.json: 100% ██████████ 53.0/53.0 [00:00<00:00, 2.45kB/s]
config.json: 100% ██████████ 612/612 [00:00<00:00, 25.0kB/s]
model.safetensors: 100% ██████████ 90.9M/90.9M [00:00<00:00, 119MB/s]
tokenizer_config.json: 100% ██████████ 350/350 [00:00<00:00, 19.1kB/s]
vocab.txt: 100% ██████████ 232k/232k [00:00<00:00, 3.58MB/s]
tokenizer.json: 100% ██████████ 466k/466k [00:00<00:00, 9.13MB/s]
special_tokens_map.json: 100% ██████████ 112/112 [00:00<00:00, 5.16kB/s]
1_Pooling/config.json: 100% ██████████ 190/190 [00:00<00:00, 8.70kB/s]
Embeddings shape: (100, 384)

```

3. Citation Regressor

3.1 Regression Model

Training Data:

The entire dataset (abstracts, titles, and publication year) was preprocessed, and text embeddings were computed using a SentenceTransformer model. The publication year was normalized and appended as an additional feature.

Model Selection:

We used GridSearchCV to tune a regression model (e.g., a RandomForestRegressor) to predict the citation count. The best model and hyperparameters were selected based on performance metrics on a train/validation split.

Fig 3.1: best regression model and hyperparameters using AutoML or GridSearch

```
Best parameters: {'max_depth': None, 'min_samples_split': 2, 'n_estimators': 50}
Best CV score (negative MSE): -3999.5312600000007
Validation RMSE: 188.00337566118327
Validation MAE: 72.569
Validation R2: 0.09753748567611786
```

Testing:

The crawled data served as the test set. The best model was applied to estimate citation counts on the test set, and regression metrics (RMSE, MAE, R^2) were reported. We discussed challenges such as the time gap between training and test data and the introduction of new research topics and authors over time.

Fig. 3.2: Test by crawled data

```
CSV files found: ['Diffusion_Models.csv', 'Foundation_Models.csv', 'Generative_Models.csv', 'LLM.csv', 'VLM.csv']
Loaded /content/drive/My Drive/extracted_crawldata/Diffusion_Models.csv
Loaded /content/drive/My Drive/extracted_crawldata/Foundation_Models.csv
Loaded /content/drive/My Drive/extracted_crawldata/Generative_Models.csv
Loaded /content/drive/My Drive/extracted_crawldata/LLM.csv
Loaded /content/drive/My Drive/extracted_crawldata/VLM.csv
Combined crawled data shape: (2350, 6)
Test RMSE: 2671.131527605483
Test MAE: 1216.938
Test R2: -0.25841791623472754
```

3.2 Bonus: Author-Author Network with a GNN

GNN Approach:

We built an author–author network from our training data and converted it into a PyTorch Geometric Data object.

Model Definition:

A Graph Neural Network (GNN) model was defined (using GCN layers) to update author node features. For each paper, the embeddings of its authors were aggregated (via mean pooling) and fed into a regression head to predict citation counts.

Training and Prediction:

The model was trained on a subset of papers, and its performance was evaluated as a bonus approach for citation regression.

Fig. 3.3: Author–Author Network

```
Built Author-Author Network from 1000-sample:
Number of nodes (authors): 3104
Number of edges (co-authorships): 4866
Converted to PyTorch Geometric Data object.
GNN model initialized.
Epoch 0, Loss: 7329.7686
Epoch 10, Loss: 7028.6768
Epoch 20, Loss: 6262.1846
Epoch 30, Loss: 6114.9224
Epoch 40, Loss: 5955.6245
Epoch 50, Loss: 5880.3252
Epoch 60, Loss: 5835.3325
Epoch 70, Loss: 5795.9004
Epoch 80, Loss: 5762.2197
Epoch 90, Loss: 5731.8008
Training complete.
Predicted citation counts for the first 10 papers:
tensor([[29.0914],
        [ 8.4136],
        [33.4840],
        [39.7248],
        [48.8853],
        [50.2984],
        [18.8971],
        [23.6434],
        [15.7983],
        [39.5838]])
```

4. Product: Research Assistant

4.1 Retrieval-Augmented Generation (RAG) Pipeline

Our Research Assistant uses a RAG-based approach to retrieve relevant papers from a vector database (built using FAISS) and to generate analytical reports. The system works as follows:

Paper Retrieval:

A 1000-paper sample from the combined dataset is embedded (using SentenceTransformer) and indexed in FAISS.

A user submits a topic query. KeyBERT is used to extract keywords from the query to enhance it.

The enhanced query is embedded and used to search the FAISS index via cosine similarity (inner product on normalized embeddings) to retrieve the topk relevant papers.

Author Query:

Users can also query based on authors. spaCy’s Named Entity Recognition (NER) extracts potential author names from the query, and the system returns papers whose authors match the extracted names.

Fig 4.1: research query

```
Enter your research query: LLM
Top relevant papers for the query:
Title: A Proxy Creation Service for SWP to XML Translations.
Abstract: No abstract available ...
-----
Title: A Pragmatic and Musically Pleasing Production System for Sonic Events
Abstract: We describe a novel application for sonic events namely their generation via mathematical functions implemented on a universal all purpose Java platform. Their design is driven by a set of requirements that arise in rec...
-----
Title: Using Imputation Techniques to Help Learn Accurate Classifiers
Abstract: It is difficult to learn good classifiers when training data is missing attribute values. Conventional techniques for dealing with such omissions, such as mean imputation, generally do not significantly improve the perfor...
-----
Title: Decoding spike train ensembles: tracking a moving stimulus
Abstract: We consider the issue of how to read out the information from nonstationary spike train ensembles. Based on the theory of censored data in statistics, we propose a 'censored' maximum likelihood estimator (MLE) for decod...
-----
Title: Identitätsmanagement für Hybrid-Cloud-Umgebungen an Hochschulen-Erfahrungen im Münchner Wissenschaftsnetz.
Abstract: No abstract available ...
```

LLM Report Generation (Bonus):

Using an LLM text generation model (via the OpenAI API, for example), the system generates a concise summary report of the retrieved papers.

The prompt is engineered to instruct the LLM to act as a research assistant that summarizes key findings, common topics, and emerging trends.

Fig. 4.2: Extracted author names

```
Extracted author names: ['Yang']
Papers matching the author query:
Title: Speech Reconstruction from human auditory cortex with deep neural networks.
Authors: ['Hinda Yang', 'Sander A. Sheth', 'Catherine A. Schevon', 'Guy H. McHann', 'Nima Mesgarani']
Abstract: No abstract available ...
-----
Title: File Transfer for Mobile Devices in Heterogeneous Radio Networks
Authors: ['Chih-Mei Yi', 'Shao-Shuan Yang', 'Yi-Bing Lin', 'Yi-Ta Chuang', 'Pin-Chuan Lin']
Abstract: In the past, although mobile devices were equipped with multiple radio interfaces, for the sake of power saving, only one was activated for data transmission. The idea of concurrent transmission via multiple radio interf...
-----
Title: Securely Secure Reliability Computation on Encrypted Location Check in Data
Authors: ['Hanyu Qian', 'Boyang Wang', 'Franklin Li', 'Ming Li', 'Yuefeng Zhang']
Abstract: No abstract available ...
-----
Title: A Public Vehicle System with Multiple Origin-Destination Pairs on Traffic Networks
Authors: ['Hong Zhu', 'Liguo Wang', 'Xiao-Yang Liu', 'Huihui Shen', 'Jin-Shu', 'Jin-Yue Ma']
Abstract: Substantial technology advances have been made in areas of autonomous and connected vehicles, which opens a wide landscape for future transportation systems. We propose a new type of transportation system, Public Vehicle...
-----
Title: A fast and accurate tracking algorithm of left ventricles in 3D echocardiography
Authors: ['Lin Yang', 'B. Georgescu', 'Yuefeng Zhang', 'David J. Foran', 'Dorin Comaniciu']
Abstract: Tracking of left ventricles in 3D echocardiography is a challenging topic because of the poor quality of ultrasound images and the speed consideration. In this paper, a fast and accurate learning based 3D tracking algorit...
-----
Title: Modelling software fault management with runtime verification
Authors: ['Xingjun Zhang', 'Yan Yang', 'Endong Wang', 'Ilseu You', 'Xiaohu Dong']
Abstract: To achieve the software fault tolerance at runtime, based on runtime verification techniques, this paper proposes a runtime model of running program, which is used to define the actions and constrains for runtime software...
```

Conclusion

In this project, we developed a comprehensive AI-based Research Assistant that:

Crawls and preprocesses scholarly article data from multiple sources.

Performs exploratory data analysis and network analysis to extract insights on publication trends, citation networks, and collaboration patterns.

Uses clustering and community detection to group similar authors and papers, and to extract keywords that represent emerging research topics.

Predicts citation counts using both traditional regression models and a bonus GNN-based approach that leverages author–author relationships.

Provides a user-facing product that supports topic-based and author-based retrieval via a RAG pipeline, enhanced by an LLM that generates concise summary reports of relevant papers.

The complete source code for each step has been provided, and all parts are integrated into a single product. This system demonstrates a full pipeline from data crawling and analysis to product development, providing an end-to-end solution for researchers to efficiently locate and analyze scholarly literature.